

A⁵OPUBLIC EVIDENCE
SURFACE

● OFFLINE-CAPABLE VERIFIER



ONEBEHALFOF™ verification of A⁵O appointment receipts: confirm who appointed the agent, on whose behalf it acted, and whether that appointment stood at the moment of execution.

The body, published keys, and dual signatures agree. This establishes integrity: any alteration to the receipt is detectable. It does not by itself establish authenticity — see Trust Anchor below.

Verify with ONEBEHALFOF™ ↗

See the trust boundary ↓

PUBLIC INPUTS

Receipt · public keys ·
appointment evidence

NEVER TRANSMITTED

Private keys · signer ·
decision logic

LIVE CONFORMANCE TEST

In-browser verification console

This is the live public verifier for
the A°O Gate at
theapertures.app. Every check
runs in this browser from public
evidence alone.

01 Authentic artifact	02 Alter principal	03 Swap published key	04 Revoke appointment	05 Reorder only
RECEIPT ARTIFACT A5O-RCT-1.4 DECISION ALLOW^{F-STOP}_{f/1.2}			PUBLIC VERDICT ONEBEHALFOF™ verified The body, published keys, and dual signatures a establishes integrity: any alteration to the receipt	

RECEIPT

RCT-A5O-PUBLIC-
DEMO-2026-07-10-
0001

APPOINTER

did:a5o:appointer:ag
ent-obo-inc

ONEBEHALFOF

did:a5o:principal:visi
onary-member-0710

AGENT

did:a5o:agent:iris-01

ACTION

book_private_aviatio
n · TEB-NCE

RECEIPT HASH

62e03d2cee69c50
bf2...
3218ce9dac3b

INSPECT
JSON

DOWNLOAD
ARTIFACT

detectable. It does not by itself establish authen
see Trust Anchor below.

✓

RECEIPT BODY

A5O-RCT-1.4 structure

All 13 fields of the hashed body are
declared-typed. The seal — receipt
signatures — sits outside the hash
construction.

✓

RECEIPT BODY

Canonical field order

onebehalf of occupies position 7 in
body.

✓

RECEIPT BODY

Receipt hash

62e03d2cee69...ce9dac3b recom
the body.

✓

PUBLISHED KEYS

Published keyset digest

7a9ef5fe592a...21eca42b matche
public record.

✓

PUBLISHED KEYS

Receipt keys are published

Both embedded verification keys n
entries in the published keyset.

✓

DUAL SEAL

Ed25519 signature

Verified independently over the UT
receiptHash message.

✓

DUAL SEAL

ML-DSA-87 signature

FIPS 204 Category 5 signature ver
same message.

✓

AUTHORITY EVIDENCE

Appointment reference

d508391b4b27...fcf5c61c is boun
body. The hash commits to an app
its contents are not disclosed and
recoverable from this receipt.

✓

AUTHORITY EVIDENCE

Principal binding

Appointer, agent and onebehalf of
distinct, matching roles.

✓

AUTHORITY EVIDENCE

Scope and revocation

The receipt binds scope and asser epoch 42. Revocation state shown from published evidence at the tim not from a live operator query. Cur confirmation requires an external r which is not yet published. The sig includes credits:<=150. No field in records the credit amount of the ac conformance on that scope elemei witnessed by the receipt.

LOCAL VERIFICATION TIME

AUTHENTICITY BOUNDARY

Trust Anchor

This verifier runs entirely in your browser and never contacts the operator. Every input it consumes — receipt, published keyset, appointment evidence — is nonetheless served from this origin.

A passing verdict therefore establishes **integrity**: the artifact is internally consistent and any alteration is detectable. It does not by itself establish **authenticity**, which requires binding the published keyset to an anchor outside this operator's control.

KEYSET DIGEST 7a9ef5fe592a...21eca42b

EXTERNAL ANCHOR Not yet published

VERIFIER SOURCE [Inspect inline source](#)

EVIDENCE, NOT ASSERTION

Inspect every public input

The seal stays portable. The verifier does not call the operator or replay the private gate.

Receipt	receipt.jsonPUBLIC · READ ONLY
Published keyset	
Appointment	
Canonical body	<pre>{ "receipt_id": "RCT-A5O-PUBLIC-DEMO-2026-07-10-0001", "schema_version": "A5O-RCT-1.4", "timestamp": "2026-07-10T14:08:00Z", "appointer": "did:a5o:appointer:agent-obo-inc", "appointment_ref": "sha256:d508391b4b27fefae6cb089fe87d285276e125862cd9e84ac600e9fefcf5c61c", "agent": "did:a5o:agent:iris-01", "onebehalf": "did:a5o:principal:visionary-member-0710", "action": { "descriptor": "book_private_aviation", "target": "TEB-NCE" }, "scope": [</pre>

```
"aviation.booking",  
"route:TEB-NCE",  
"credits:<=150"  
],  
"decision": "ALLOW",  
"reason_code": "RC-OK-VISIONARY-  
AVIATION".
```

Canonicalization — normative

A5O-RCT-1.4 canonical form. The hashed body is the receipt object with `receiptHash` and `signatures` removed, serialized as JSON with keys in the declared canonical order — **not sorted** — with separators `,` and `:`, UTF-8 encoded, with no whitespace, no prefix, no suffix and no trailing newline. For this receipt the preimage is 570 bytes.

`receiptHash` is the SHA-256 digest of that byte string, lowercase hexadecimal.

The signature message for **both** suites is the 64-character lowercase ASCII-hex encoding of that digest. The FIPS 204 context string is empty.

Note for implementers: sorted-key canonicalization (RFC 8785 / JCS) produces a different digest and will fail a valid receipt.

Verifier source — hashing and signature checks

```
const canonicalBody = JSON.stringify(body);  
const calculatedHash = sha256(  
  new TextEncoder().encode(canonicalBody)  
);  
const message = new TextEncoder().encode(calculatedHash);  
  
ed25519.verify(signature, message, publicKey, { zip215: false  
ml_dsa87.verify(signature, message, publicKey);
```

The running page uses @noble/hashes, @noble/curves, and @noble/post-quantum for these checks. Scenario results are produced by the same verification path.

TRUST BOUNDARY

The proof travels. The secrets do not.

PUBLIC VERIFIER

01

Receives evidence

Signed receipt artifact

Published public
keyset

Appointment +
revocation evidence

ONE-WAY RELEASE



public material only

OPERATOR

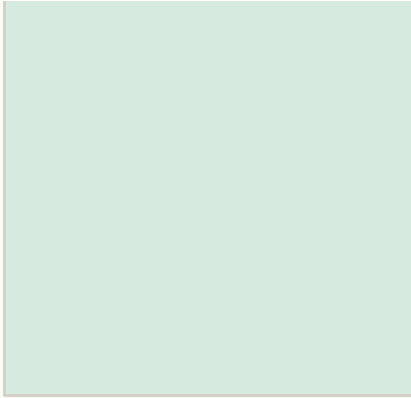
02

BOUNDARY

Retains control

Ed25519 + ML-
DSA-87 private
keys

Gate and policy
engine



Signing ceremony
+ runtime state

PRIVATE
MATERIAL
NEVER SHIPPED

THE LOAD-BEARING DISTINCTION

*“An unbound
principal
field is a
label, not a
proof.”*

In RCT 1.4, onebehalfof sits inside the canonical receipt body. Change the principal and the hash changes. Change the hash and both signatures fail.

A⁵O

Live verifier. CANON-SIGN 2.0 and RCT 1.4 are presented as the author's draft, pending Institute ratification. No production private key is present in this site.

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<https://verify.theapertures.app>

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PUBLIC AUTHORITY EVIDENCE

eSigned on **LegalZoom** 62.215.38.249

Fitzgerald

09:29:45 GMT

J. Heslop

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